TIANCHONG JIANG

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Google Scholar Chicago, IL

Education

Toyota Technological Institute at Chicago Sep. 2024 – Jun. 2028 (Expected)
PhD in Computer Science
Advisor: Matthew R. Walter

University of Chicago Sep. 2019 – Dec. 2023
B.A. Mathematics | B.S. Computer Science with Honors
GPA: 3.89/4.00; magna cum laude

Publications and Preprints

T. Jiang*, R. Xu*, Y. Li, X. Tan, and M. R. Walter. "Real-to-Sim-to-Real at the Last Millimeter," in *Conference on Robot Learning (CoRL) Late-Breaking Results*, 2026.

Ludo Robotics (incl. **T. Jiang**). "Ludi$_{0.1}$: An Agentic System for Socially Intelligent Robots," *Technical report, arXiv:2608.22035*, 2026.

T. Jiang, X. Tan*, S. W. Wheeler*, L. Sun*, T. W. Ayalew*, and M. R. Walter. "What Are We Actually Benchmarking in Robot Manipulation?," in *Conference on Robot Learning (CoRL)*, 2026. **Oral, Best Paper Award Finalist**.
Also at the *IROS 2026 RGMCW Workshop*: **RGMCW Workshop Best Abstract Award**.

T. Jiang, J. Ji, X. Tan, J. Fang, A. Bhattad, V. Guizilini†, and M. R. Walter†. "Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026. **Best Paper Award on Robot Learning**.

X. Tan, J. Ji, **T. Jiang**, P. Lopes, and M. R. Walter. "HapCompass: A Rotational Haptic Device for Contact-Rich Robotic Teleoperation," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

L. Sun*, T. Yoneda*, S. W. Wheeler, **T. Jiang**, and M. R. Walter. "StackGen: Generating Stable Structures from Silhouettes via Diffusion," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

T. Ayalew, X. Zhang*, K. Y. Wu*, **T. Jiang**, M. Maire, and M. R. Walter. "PROGRESSOR: A Perceptually Guided Reward Estimator with Self-Supervised Online Refinement," in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

T. Yoneda*, J. Fang*, P. Li*, H. Zhang*, **T. Jiang**, B. Picker, D. Yunis, S. Lin, H. Mei, and M. R. Walter. "Statler: STATe-maintaining Language models for Embodied Reasoning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

T. Yoneda*, **T. Jiang***, G. Shakhnarovich, and M. R. Walter. "6-DoF Stability Field via Diffusion Models," *arXiv preprint*, 2023.

Experience

AI/ML Research Intern, Ludo Robotics / KRAFTON June 2026 – August 2026
• Worked on sim-to-real transfer for a humanoid robot, including reinforcement learning (PPO) teacher policies trained in Isaac Sim.
• Cross-group, cross-culture collaboration between the US simulation team and the Korea (Seoul) real-robot team.

Research Assistant, Robot Intelligence through Perception Lab October 2024 – Present
• Studying view-invariant imitation learning with camera conditioning and Plucker ray representations for robust visuomotor policies.
• Auditing robot manipulation benchmarks for shortcut solvability, statistical fragility, creeping overfitting, and data-source dependence.
• Collaborated with Toyota Research Institute and the UChicago Human Computer Integration Lab.

Research Assistant, Robot Intelligence through Perception Lab August 2022 – June 2024

- Implemented an SE(3) diffusion model to predict stable object configurations. (MuJoCo, PyTorch)
- Co-developed a Large Language Model interface for a UR5 robot arm, demonstrated to thousands of visitors at the Museum of Science and Industry in Chicago. (ROS, UR5)

Teaching

TA for MPCS 50103 Discrete Mathematics Autumn 2022

TA for TTIC 31170 Planning, Learning, and Estimation for Robotics and Artificial Intelligence Spring 2025

Service

Reviewer for IROS 2025; ICCV 2025; ICRA 2026; CoRL 2026; ACL ARR 2026; AAAI 2027; TMLR

Skills

Python, PyTorch, ROS, MuJoCo, Isaac Sim, Docker, LaTeX, Git, Slurm

Robot Platforms: UR5, Unitree G1, Hiwonder NexArm, Koch Arms, SO-101 Arms, ARCTOS, TurtleBot

Relevant Coursework

Introduction to Machine Learning; Algorithms; Algorithmic Game Theory; Fundamentals of Deep Learning; Generative Models, Art and Perception; Planning, Learning, and Estimation for Robotics and Artificial Intelligence; Unsupervised Learning; Mathematical Toolkit; Learning on Graphs and Manifolds; Introduction to Computer Vision; Introduction to Statistical Machine Learning